

ANDREAS MENTZELOPOULOS

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EDUCATION

Massachusetts Institute of Technology - MIT	2020 – 2025
PhD Mechanical Engineering and Computation, minor in Finance	
SM Mechanical Engineering	2020 – 2022
University of Michigan – Ann Arbor	2016 – 2020
BSE Mechanical Engineering, minor in Mathematics	
BSE Naval Architecture & Marine Engineering	

INTERESTS & BIO

Generative Modeling, Time-series Forecasting, Deep Learning

I am a quantitative researcher at [Cubist Systematic Strategies](#) (Point72) working on equity [statistical arbitrage](#) trading and research. Before Wall Street, I was a ML scientist at [Wayfair](#) working on the measurement & attribution team, developing [multi-touch attribution](#) models by leveraging explainable deep-learning methods (Deep Nets + Shapley). My work helped allocate more than 1 billion USD in annual marketing spend. I completed my PhD at MIT, where I developed digital twins for marine risers. I pioneered DL architectures for [time-series forecasting](#) and [simulation methods using generative modeling](#) (see Chapter 8). I also co-founded and led [LOBSTgER](#). I have interned for MathWorks, CITIC Securities CLSA, and the American Bureau of Shipping.

SKILLS

Computer languages	Python, MATLAB, C++ (familiar)
Machine/Deep Learning frameworks	PyTorch, TensorFlow, scikit-learn
Developed gen-AI models	Diffusion, VAE, GANs
Developed time-series models	Transformers, LSTM, GRU, XGBoost, LightGBM

WORK EXPERIENCE

Research Analyst (Quantitative Researcher), Cubist Systematic Strategies (Point72), NY, NY	06/2026 – pres.
- Equity statistical arbitrage quantitative trading & research (Point72). Trading team of 7.	
Wayfair, Marketing & Measurement Science, Machine Learning Scientist, Boston, MA	08/2025 – 06/2026
- Multi-touch attribution modeling with explainable deep-learning (Deep Nets + Shapley).	
- Marketing lift studies (conversion & geographical).	
CITIC Securities CLSA, Quantitative Trading Strategies, Intern, New York, NY	06 – 08/2024
- Equity statistical arbitrage research (Asia-pacific alternative markets).	
MathWorks, Engineering Development Group, Intern, Natick, MA	06 – 08/2023
- Quality engineering for Simscape multibody, Gas, Moist Air, and Isothermal Liquid libraries.	
- Design & implementation of customer facing forklift example model , which was publicly released with MATLAB R2024a.	
American Bureau of Shipping, Offshore Equipment Group, Intern, Houston, TX	06 – 08/2020
- Engineering reviews for pressure vessels as per ASME Section VIII Div. 1 (high-stakes offshore and marine environments)	
- Classification of BP’s Mad Dog Phase 2 – Argos semi-sub drilling platform.	
- Classification of 12 chemical tankers per IBC Code, Jan. 21.	

RESEARCH EXPERIENCE

Stochastic Analysis and Nonlinear Dynamics Lab, Tow Tank, MIT

09/2020 – 09/2025

Graduate Student Research Assistant, PI: Prof. Michael Triantafyllou, Prof. Themis Sapsis

1. LOBSTgER (Learning Oceanic Bioecological Systems Through generative Representations):

- Developed a family of diffusion models to synthesize and enhance high-resolution awareness photography images, focusing on marine animals local to Massachusetts and the Gulf of Maine. Project in collaboration with photographer [Keith Ellenbogen](#), who provides training data.
- Developed unconditional latent diffusion models generating 640x1024 images, which generalize to higher resolutions. Please find samples [here](#).
- Developed conditional latent diffusion models for underwater image restoration and enhancement tasks, particularly focusing on low-visibility, color distortion, and lens blocking issues of underwater photography. Please find samples [here](#).
- Recognized on [MIT News](#) (06/25). Recognized at the “Meet the Scientists Series” (10/24) and “Computer Science Education Weekend” (12/24) at the Museum of Science in Boston.

2. *Firstling* Digital Twin (for marine risers):

- Developed a transformer-based deep neural network for underwater pipeline vibrations forecasting.
- The twin predicts the pipeline’s full motion (nonlinear, nonstationary dynamics) from sparse measurements on the body. The model accommodates wave input data if available and can forecast fatigue.
- The model has been validated on field data from BP’s Atlantis platform operating in the Gulf of America.

3. Virtual Towing Tank (for vortex-induced vibrations):

- Conducted studies to determine the most suitable generative model for pipe vibration data, comparing GANs, VAEs, and Diffusion models.
- Developed conditional latent diffusion models that generate densely sampled time-series based on physical parameters like tension and flow velocity as conditioning information, equivalent to altering experimental conditions in the lab.
- Validated on data collected at the MIT Towing Tank.
- Generated data are higher fidelity than CFD data and are obtained much faster with significantly less compute.

SELECTED PUBLICATIONS ----- (for full list reference [google scholar](#), pre-prints [here](#))

1. **(Submitted) Mentzelopoulos, A.,** Fan, D., Sapsis, T., Karniadakis, G., Triantafyllou, MS, “[Deep generative modeling of vortex-induced vibrations](#)” (pre-print), Engineering Applications of Artificial Intelligence, 2025.
2. **Mentzelopoulos, A.,** Fan, D., Sapsis, T., Triantafyllou, MS, “[Deep learning vortex-induced vibrations: time-space forecasting with transformers](#)”, Journal of Fluids and Structures, 2025.
3. **Mentzelopoulos, A.,** Fan, D., Sapsis, T., Triantafyllou, MS, “[Variational autoencoders and transformers for multivariate time-series generative modeling and forecasting: Applications to vortex-induced vibrations](#)”, Ocean Engineering, 2024.
4. **Mentzelopoulos, A.** Ferrandis, J.d.A., Rudy, S., Sapsis, T., Triantafyllou, M.S, Fan, D., “[Data driven prediction and study of vortex induced vibrations by leveraging hydrodynamic coefficient databases learned from sparse sensors](#)”, Ocean Engineering, 2022.

SELECTED HONORS & AWARDS ----- (for full list reference [personal website](#))

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| 1. Onassis Foundation Scholarship , Onassis Foundation (most prestigious graduate scholarship awarded to Greek students globally) | 12/2022 |
| 2. MathWorks Fellowship , MathWorks (awarded to 20 MIT students) | 09/2022 |
| 3. Society of Naval Architecture and Marine Engineering Award , MIT (awarded to 1 student at MIT) | 05/2021 |
| 4. William M. Kennedy Scholarship , Society of Naval Architects & Marine Engineers (awarded to 1 student globally) | 04/2021 |
| 5. MIT SMA2 Fellowship , MIT (awarded to 1 graduate student at MIT) | 09/2020 |
| 6. Undergraduate Scholarship , Society of Naval Architects & Marine Engineers (awarded to 4 students globally) | 07/2019 |
| 7. ABS Scholarship , American Bureau of Shipping | 05/2018 |

RELEVANT COURSEWORK (MIT)

Machine Learning

1. Machine Learning (6.7900)
2. Deep Learning (6.S898)
3. Parallel Computing & Scientific Machine Learning (6.7320)
4. Computer Vision (6.8300)
5. Nonlinear Optimization (6.7220)
6. Intro to Machine Learning (6.036)

Finance

1. Advanced Data Analytics and Machine Learning in Finance (Natural language processing with applications in Finance - 15.S08)
2. Financial Engineering (15.456)
3. Managerial Finance (15.041)

Mathematics

1. Numerical Methods for Partial Differential Equations (6.7330)
2. Stochastic Systems (2.122)
3. Numerical Fluid Mechanics (2.29)
4. Dynamics (2.032)
5. Marine Hydrodynamics (2.20)